09/03

cnt

5

0-4 is valid index

cart[0]
cart[1]
cart[2]
cart[3]
cart[4]

book	100	5	cart
pen	55	10	cart+1
pencil	10	20	cart+2
eraser	5	5	cart+3
scale	10	2	cart+4

① delete by product name

(a) Search the product

(b) if found perform memmove

(c) else say not found

5-2 = 3

5-2-1 = 2 * 28 → 56

memmove(cart+i, cart+i+1, (cnt-i-1)*sizeof(PRODUCT));

cart[0]
cart[1]
cart[2]
cart[3]
cart[4]

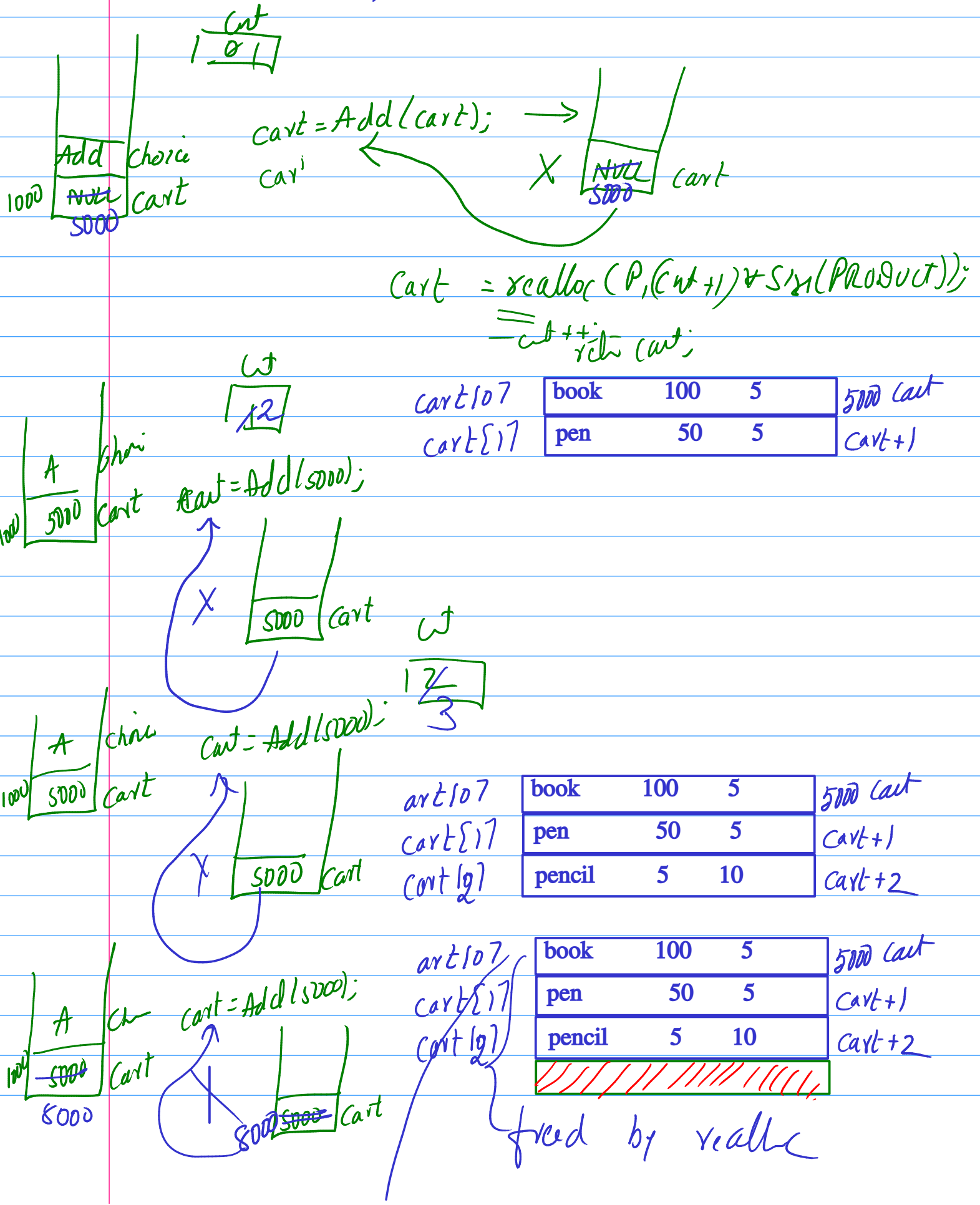
book	100	5	cart
pen	55	10	cart+1
eraser	5	5	
scale	10	2	

cnt--;

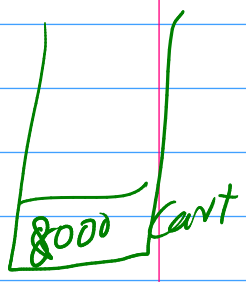
cnt
4

PRODUCT cart with DMA

PRODUCT *cart=NULL;



data copied to new location



cart[0]	book	100	5
cart[1]	pen	50	5
cart[2]	pencil	5	10
cart[3]	eraser	5	5

cart
8000
cart+1
cart+2
cart+3

Del based on product name

prod[]="pencil"

cart[0]	book	100	5
cart[1]	pen	50	5
cart[2]	pencil	5	10
cart[3]	eraser	5	5
cart[4]	scaler	10	2
cart[5]	sharpner	20	2

cart
8000
cart+1
cart+2
cart+3
cart+4
cart+5

```
for(i=0; i<cnt ; j++)
{ if(strcmp(prod, cart[i].name)==0)
  {
  memmove(cart+i, cart+i+1, (cnt-i-1)*sizeof(PRODUCT));
```

cart[0]	book	100	5
cart[1]	pen	50	5
cart[2]	eraser	5	5
cart[3]	scale	10	2
cart[4]	sharpner	20	2
cart[5]	sharpner	20	2

a
8000
cart+1
cart+2
cart+3
cart+4
cart+5

ct
6
5

cnt--;

cart = realloc(cart, cnt * sizeof(PRODUCT));

ct

5

arr[0]	book	100	5	a 8000
arr[1]	pen	50	5	cart+1
cart[2]	eraser	5	5	cart+2
cart[3]	scale	10	2	cart+3
cart[4]	sharpner	20	2	cart+4